

Period 4 compounds as catalysts

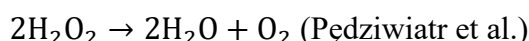
How does the increase in electron number for period 4 ions affect its efficacy as a catalyst in the decomposition of Hydrogen Peroxide?

Introduction

Catalysts are a very important substance in chemistry, as they can reduce the amount of energy needed for a reaction to happen. As the global community is more and more concerned about problems such as global warming and climate change, reducing the ecological footprint through lessening the amount of energy we consume is a critical goal we need to achieve. One solution we could do is using better catalysts in industry to increase the rate of reaction and reducing the heat needed to produce something. As such, this experiment would be conducted to learn more about this phenomenon to see which catalysts are better in the Period 4 elements.

Background

Hydrogen peroxide(H_2O_2) is a peroxide; a compound that has an oxygen-oxygen bond. Both oxygen atoms have an oxidation state of -1, while other compounds containing oxygen have an oxidation state of -2 or 0, causing it to tend to decompose in a pure form to create water (H_2O) and oxygen (O_2). The formula for the decomposition is as follows:



A catalyst a substance that alters the rate of a chemical reaction but is itself unchanged at the end of the reaction. (Concise Medical Dictionary) Catalysis is undergone in the active site, a region on the surface of the catalyst. As such, the larger the catalyst, the larger the active site, and the higher the rate of reaction as more reactants can be in contact with the active site.

There are 2 kinds of catalysts: heterogeneous and homogeneous catalysts. A heterogeneous catalyst has a different state than the reactants, such as when the catalyst is in a solid state and the reactants are in a liquid state. A homogeneous catalyst has the same state as the reactant mixture, such as when both the catalyst and the reactants are in a liquid state.

In heterogeneous catalysis, the molecules of the reactants are adsorbed onto the surface of the catalyst, weakening the bonds of the molecules. When another reactant molecule is attached or collides with the catalyst, a reaction between them happens, after which the molecules are desorbed. The efficacy of the catalyst is determined by the surface area of the catalyst; once the active of the catalyst is fully filled by reactants, the reaction is interrupted until the molecules on the surface react and desorb.

In homogeneous catalysis, the molecules of the catalyst are uniformly spread among the reactant mixture, allowing them to mix. This causes the number of interactions between the catalyst and the reactant per unit time to be much higher than heterogeneous catalysis, as there are more active sites. This is because the molecules of the reactants and catalysts can mix together, thus more molecules of the catalyst can interact with the molecules of the reactants. However, one downside is that the recovery of homogeneous catalysts is harder as the use of techniques such as filtration or distillation is needed.

As such, my research question is, which substance is most effective at catalysing the decomposition of 3% hydrogen peroxide in room temperature and neutral pH, Iron (III) Chloride (FeCl_3), Copper (II) Chloride (CuCl_2), Manganese (IV) Oxide (MnO_2), Potassium Manganate (VII) (KMnO_4) and Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)?

Hypothesis

The hypothesis is that Potassium Manganate (VII) would be the best at catalysing Hydrogen Peroxide, giving the fastest rate of reaction.

All these substances have a transition metal ion in their molecules, which have a partially filled d-orbital, allowing them to take or give electrons. They can be in and change between

different oxidation states, in addition to their ability to form complexes, making them good catalysts, due to being able to form unstable intermediate compounds with the help of an inconsistent oxidation state (*Catalytic properties of transition metals*). These compounds can create new roads for the decomposition by reducing the amount of activation energy needed for the reaction to take place. Except for Manganese (IV) Oxide, all of them are soluble, so they can mix with the Hydrogen Peroxide and allow more successful collisions to happen.

As, Iron (III) Chloride and Copper (II) Chloride are homogeneous catalysts, there is an assumption that these substances would have a higher rate of reaction than the other substances. However, as iron commonly is in the +2 or +3 oxidation state, and Copper in the +1 or +2 oxidation state, its ability to form complexes is limited compared to Chromium and Manganese, which has +2, +3, +6 and +2, +3, +4, +6 and +7 oxidation states (Libretexts), which increases its ability to form different complexes, with Manganese being the best. This is because their ability to receive and donate more electrons are much better with a higher range of oxidation states.

However, Potassium Manganate's reaction with Hydrogen Peroxide is autocatalytic, as the product of reaction Mn^{2+} will create an unstable complex. Due to its instability, it reacts with the Hydrogen Peroxide, catalysing it, increasing the rate of reaction. (Pędziwiatr et al.)

Variables

The variables are as follows:

Variable type	Variable	Method of measurement/control (Uncertainty in measurements)	Effect of variable on experiment
Independent Variable	Time taken for the Hydrogen Peroxide to fully decompose	Stopwatch ($\pm 0.01\text{s}$)	Using a stopwatch would cause random errors due to human reaction

Dependent Variable	The type of catalyst used; Iron (III) Chloride (FeCl_3), Copper (II) Chloride (CuCl_2), Manganese (IV) Oxide (MnO_2), Potassium Manganate (VII) (KMnO_4) and Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) these are discontinuous variables		
Controlled Variable	Concentration of H_2O_2 (3.000%)	Preparation and calculation ($\pm 0.003\%$)	Will be constant throughout the experiment as concentration is proportional to rate of reaction
	Volume of H_2O_2 ($5\text{cm}^3 \pm 0.1\text{cm}^3$)	Volumetric flask ($\pm 0.1\text{cm}^3$)	Will be constant throughout the experiment as volume is proportional to rate of reaction
	Amount of catalyst used (0.01 Mol)	Electronic balance ($\pm 0.001\text{g}$) The moles of the catalyst are calculated through	Will be constant throughout the experiment as the number of particles will affect the surface area of the catalyst, which is

		$\text{Moles} = \frac{\text{Mass}}{\text{Molar Mass}}$	proportional to rate of reaction
	Form of catalyst utilised	The catalysts are used are in an anhydrous form and are crushed finely using a mortar and pestle	
Uncontrolled variables	Powdered catalysts sticking together Other chemicals used in stabilising H_2O_2 Catalyst poisoning Surface area of catalysts in powder form Time taken to seal the boiling tube		

Experimental Setup

List of chemicals

- Iron (III) Chloride $1.62\text{g} \pm 0.01\text{g}$
- Copper (II) Chloride $1.34\text{g} \pm 0.01\text{g}$
- Manganese (IV) Oxide $0.87\text{ g} \pm 0.01\text{g}$
- Potassium Manganate (VII) $1.58\text{g} \pm 0.001\text{g}$
- Potassium Dichromate $2.94\text{ g} \pm 0.001\text{g}$
- 30% Hydrogen Peroxide $5\text{cm}^3 \pm 0.1\text{cm}^3$

List of apparatus:

- Glass syringe $100\text{cm}^3 \pm 0.1\text{cm}^3$
- Bung
- Volumetric Flask, $500\text{cm}^3 \pm 0.1\text{cm}^3$
- Burette, $100\text{mL} \pm 0.02\text{ mL}$
- Boiling tube with a side arm
- Electronic scale, $\pm 0.01\text{g}$
- Thermometer, $100^\circ\text{C} \pm 0.1^\circ\text{C}$
- Pipette $5\text{mL} \pm 0.01\text{ mL}$
- Plastic tube
- Stand
- Clamps

Safety precautions

The chemicals used in this experiment can cause skin and eye irritation. As such, goggles and gloves should be worn when handling them and when conducting the experiments. Some of the chemicals may be harmful when swallowed, so a mask is worn and not taken off so that they are not ingested.

The decomposition of hydrogen peroxide is highly exothermic at 23.44 kcal/mol (H_2O_2 self-accelerated decomposition) and can heat up the boiling tube, so tongs are used to hold the boiling tube after the experiment is conducted.

The rate of reaction of Hydrogen Peroxide when a catalyst is added is measured using a stopwatch starting from the time when the catalyst is put into the hydrogen peroxide until all the oxygen is evolved, or when the glass syringe stops moving. This method is the fairest way to conduct the experiment. If titration is used, the removal of the catalyst must be done, but as some of the catalysts are soluble in water, this will cause a lot of other uncontrollable variables to be added. Quenching would also increase the uncontrollable variables as the hydrogen peroxide would continue to decompose even when it is not in the presence of a catalyst.

Experimental Setup

Oxygen is collected through a glass syringe. The experimental setup is as follows:

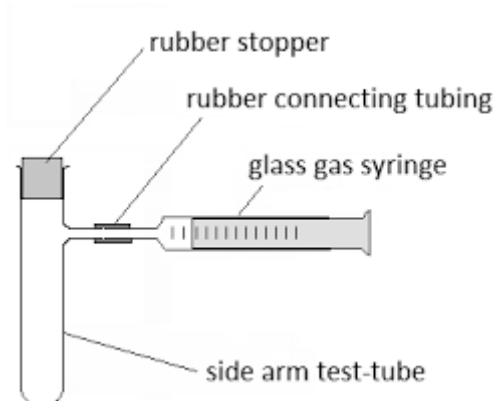


Figure 1: annotated diagram of the experimental setup

<https://www.nuffieldfoundation.org/sites/default/files/files/Rates%20of%20reaction%20-%20merged%20PDF.pdf>

Preparation of the experiment:

A $500\text{cm}^3 \pm 0.1\text{cm}^3$ 3% Hydrogen Peroxide solution is prepared from $50\text{cm}^3 \pm 0.1\text{cm}^3$ of 30% standardised hydrogen peroxide solution and $450\text{cm}^3 \pm 0.1\text{cm}^3$ of deionized water using a volumetric flask and a burette.

3% of Hydrogen Peroxide solution is selected as it would release ten times of its volume as oxygen. For 5ml of Hydrogen Peroxide, 50ml of Oxygen will be produced as a result. As the capacity of the glass syringe is only $100\text{cm}^3 \pm 0.1\text{cm}^3$, a higher concentration of Hydrogen Peroxide would cause the oxygen to overflow. As the rate of reaction would be higher as the concentration of Hydrogen Peroxide is proportional to it, so collisions would happen more frequently, accelerating the time taken for the oxygen to evolve, which would increase the inaccuracy in timing how long it takes for the reaction to finish as a timer is used to measure the time taken. A 5mL Pipette is used to extract and inject the H_2O_2 as it is the most accurate tool to measure and transfer the solution.

Before the experiment, a thermometer is used to measure the temperature of Hydrogen Peroxide to see if it is within the suitable range of temperature.

Before measuring the weight of the catalysts, the catalysts that were not in a powdered form were grounded finely using a mortar and pestle. Then the catalyst was measured and prepared for each reaction.

A glass syringe is used to measure the oxygen produced, as it is most accurate as it can give the immediate amount of volume of oxygen produced in an accurate manner compared to an inverted measuring cylinder submerged in water, as the bubbles would need to travel up the measuring cylinder. The glass syringe is secured with a clamp attached to a stand to prevent the inner syringe part from sliding out or moving before and during the experiment. The boiling tube is also secured with another clamp attached to the stand to prevent any of the chemicals from spilling.

Experimental Procedures:

1. $5.00 \text{ cm}^3 \pm 0.01 \text{ cm}^3$ of 3% H_2O_2 was withdrawn using a Pipette $5 \text{ mL} \pm 0.01 \text{ mL}$, then deposited into the boiling tube
2. The catalyst was dropped into the boiling tube and sealed using a bung and the timer is started
3. When the glass syringe stops moving, the timer is stopped, and the data is recorded
4. Repeat steps 1-3 for each catalyst for six times

Raw Data

Qualitative Data:

Observations of the experiments are listed below:

Catalyst	Observations during experiment	Observations after experiment
Iron (III) Chloride	Iron (III) Chloride is dissolved into the liquid,	The boiling tube was heated Solution became grey black residue

	turning it from transparent to a dark black colour Effervescence occurs during the experiment	
Copper (II) Chloride	Copper (II) Chloride is dissolved into the liquid, turning it from transparent to a light blue colour Small bubbles were formed	Solution remained light blue
Manganese (IV) Oxide	Iron (III) Chloride is dissolved into the liquid, turning it from transparent to a dark black colour. High amount of effervescence occurs during the experiment	The boiling tube was heated Solution remained black
Potassium Manganate (VII)	Potassium Manganate (VII) is dissolved into the liquid, turning it from transparent to a dark grey colour	The solution became dark purple The boiling tube was heated
Potassium Dichromate	Potassium Dichromate is dissolved into the liquid, turning it from transparent to orange	The solution became orange

Quantitative Data

- Iron (III) Chloride $1.62\text{g} \pm 0.01\text{g}$
- Copper (II) Chloride $1.34\text{g} \pm 0.01\text{g}$

- Manganese (IV) Oxide $0.87 \text{ g} \pm 0.01\text{g}$
- Potassium Manganate (VII) $1.58\text{g} \pm 0.001\text{g}$
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The time taken for the oxygen to fully evolve during decomposition of a 3% solution of H_2O_2 when catalysed by FeCl_3 , CuCl_2 , Ag, MnO_2 , KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ in a pH-neutral environment at room temperature is shown here

Trial (s)	FeCl_3	CuCl_2	MnO_2	KMnO_4	$\text{K}_2\text{Cr}_2\text{O}_7$
First trial	9.89	20.38	8.22	5.61	9.02
Second trial	10.01	20.25	8.23	5.84	8.98
Third trial	10.38	20.54	8.34	6.05	8.88
Fourth trial	10.26	20.58	8.09	6.08	9.12
Fifth trial	10.3	20.47	8.41	5.77	9.20
Sixth trial	9.98	20.46	8.24	5.86	9.09
Average	10.14	20.45	8.26	5.87	9.05

Figure 2: Time taken for the decomposition to take place

Analysed data

The error for each trial is $\pm 0.01\text{g}$ for the mass of the substance and $\pm 0.01\text{s}$ for the time taken for the decomposition to fully take place. As such, averaging out the values for all the trials, the average error is $\pm 0.06\text{g}$ for the mass of the substance and $\pm 0.06\text{s}$ for the time taken for the decomposition to fully take place. A graph for the data is shown below:

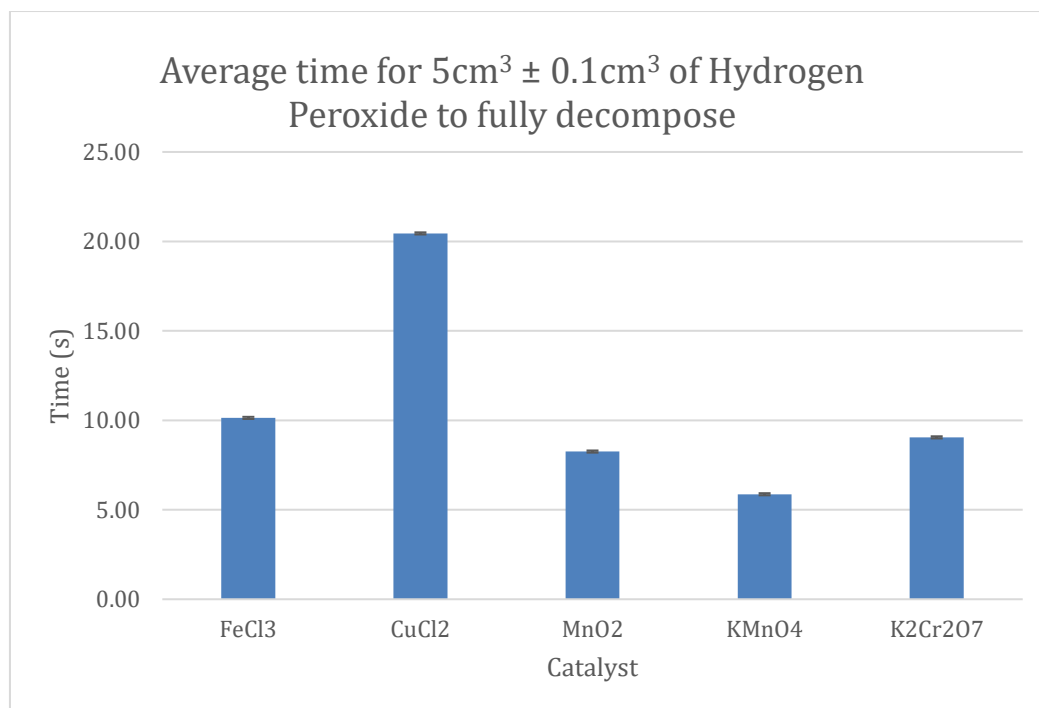


Figure 3, Average time taken for decomposition to take place

Conclusion

The results showed that among the investigated catalysts, potassium manganate (VII), which decomposed the Hydrogen Peroxide the fastest at 5.87s, was the most efficient catalyst, correlating with my hypothesis. This conclusion is consistent with findings in the literature; among the examined catalysts, KMnO_4 is the most effective at catalysing the breakdown of H_2O_2 despite the lack of quantifiable data regarding its efficacy.

The main source of uncertainty was the measurement of time, which was affected by the experimenter's reaction time as well as the tiny size of the values being measured and the low precision of the stopwatch. The result of the experiment is also by the random error related to the creation of the 3% hydrogen peroxide solution because of the decomposition of it over time.

Although some of the studied catalysts were heterogeneous and some were homogeneous, all of them were introduced directly to the solution in solid form. Due to the reduced degree to which an undissolved catalyst interacts with the reactant molecules, this may have slowed the homogeneous catalysis. However, this source of inaccuracy was not substantial for the

experiment's results since all mixtures with soluble catalysts were homogeneous at the end of the experiment.

Evaluation

Sources of error	Significance and evidence	Improvements
Different surface area of catalysts, as some of them had stuck together	A catalyst with a smaller surface area would produce a slower rate of reaction, increasing the time taken for the reaction to take place	Mix with small amounts of water so it loosens up
Catalysts not falling at the same time, some of the catalysts do not fall into solution	Initial rate of reaction is disturbed, which may extend the time taken to produce the oxygen, increasing the time taken for the reaction to take place	Put the catalyst on a flat piece of paper first before pouring in by putting the bung on it and slipping it off
Reaction time of the experimenter in stopping the stopwatch	Increases the time taken for the reaction to take place	Utilise video analysis to see when the gas syringe stops moving

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Table of data

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1st trial	9.89	20.38	8.22	5.61	9.02
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